Logistic Regression with Python - Case Study

For this case study we will work with the Titanic Data Set from Kaggle. This is a very famous dataset and very often adopted for students' first step in machine learning!

We'll be trying to predict a classification - survival or deceased - by implementing Logistic Regression in Python.

We'll use a "semi-cleaned" version of the titanic dataset; if you use the data set directly hosted on Kaggle, you may need to perform some additional data cleaning operations not shown in this lecture notebook.

Import Data Analysis and Visualization Libraries

In [35]:
```
%pip install numpy pandas matplotlib seaborn openpyxl
%pip install ipywidgets

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

```
Requirement already satisfied: numpy in /usr/local/python/3.14.2/lib/python
3.14/site-packages (2.5.4)
Requirement already satisfied: pandas in /usr/local/python/3.14.2/lib/python
3.14/site-packages (3.1.0)
Requirement already satisfied: matplotlib in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (3.11.2)
Requirement already satisfied: seaborn in /usr/local/python/3.14.2/lib/pytho
n3.14/site-packages (0.13.2)
Requirement already satisfied: openpyxl in /usr/local/python/3.14.2/lib/pyth
on3.14/site-packages (3.1.5)
Requirement already satisfied: python-dateutil>=2.9.0 in /home/codespace/.lo
cal/lib/python3.14/site-packages (from pandas) (3.0.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/python/3.14.2/
lib/python3.14/site-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/python/3.14.2/lib/
python3.14/site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in /usr/local/python/3.14.
2/lib/python3.14/site-packages (from matplotlib) (4.68.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/python/3.14.
2/lib/python3.14/site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /home/codespace/.local/li
b/python3.14/site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in /usr/local/python/3.14.2/lib/pyt
hon3.14/site-packages (from matplotlib) (13.0.0)
Requirement already satisfied: pyparsing>=3 in /usr/local/python/3.14.2/lib/
python3.14/site-packages (from matplotlib) (3.3.3)
Requirement already satisfied: et-xmlfile in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.6 in /home/codespace/.local/lib/python
3.14/site-packages (from python-dateutil>=2.9.0->pandas) (1.17.0)
Requirement already satisfied: tzdata in /home/codespace/.local/lib/python3.
14/site-packages (from python-dateutil>=2.9.0->pandas) (2026.6)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: ipywidgets in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (8.1.9)
Requirement already satisfied: comm>=0.1.3 in /home/codespace/.local/lib/pyt
hon3.14/site-packages (from ipywidgets) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in /home/codespace/.local/lib/
python3.14/site-packages (from ipywidgets) (9.17.1)
Requirement already satisfied: traitlets>=4.3.1 in /home/codespace/.local/li
b/python3.14/site-packages (from ipywidgets) (5.16.1)
Requirement already satisfied: widgetsnbextension~=4.0.16 in /usr/local/pyth
on/3.14.2/lib/python3.14/site-packages (from ipywidgets) (4.0.16)
Requirement already satisfied: jupyterlab_widgets~=3.0.17 in /usr/local/pyth
on/3.14.2/lib/python3.14/site-packages (from ipywidgets) (3.0.17)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in /home/codes
pace/.local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets)
(1.1.1)
Requirement already satisfied: jedi>=0.18.2 in /home/codespace/.local/lib/py
thon3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.20.1)
Requirement already satisfied: matplotlib-inline>=0.1.6 in /home/codespace/.
local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.2.2)
Requirement already satisfied: pexpect>4.6 in /home/codespace/.local/lib/pyt
hon3.14/site-packages (from ipython>=6.1.0->ipywidgets) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /home/codesp
ace/.local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets)
```

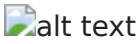
```
(3.0.53)
Requirement already satisfied: psutil>=7 in /home/codespace/.local/lib/pytho
n3.14/site-packages (from ipython>=6.1.0->ipywidgets) (7.2.2)
Requirement already satisfied: pygments>=2.14.0 in /home/codespace/.local/li
b/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (2.21.0)
Requirement already satisfied: stack_data>=0.6.0 in /home/codespace/.local/l
ib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.6.3)
Requirement already satisfied: wcwidth>=0.1.4 in /home/codespace/.local/lib/
python3.14/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0
->ipywidgets) (0.9.4)
Requirement already satisfied: parso<0.9.0,>=0.8.7 in /home/codespace/.loca
l/lib/python3.14/site-packages (from jedi>=0.18.2->ipython>=6.1.0->ipywidget
s) (0.8.7)
Requirement already satisfied: ptyprocess>=0.5 in /home/codespace/.local/li
b/python3.14/site-packages (from pexpect>4.6->ipython>=6.1.0->ipywidgets)
(0.7.0)
Requirement already satisfied: executing>=1.2.0 in /home/codespace/.local/li
b/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidge
ts) (2.3.0)
Requirement already satisfied: asttokens>=2.1.0 in /home/codespace/.local/li
b/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidge
ts) (3.0.2)
Requirement already satisfied: pure-eval in /home/codespace/.local/lib/pytho
n3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets) (0.
2.4)
Note: you may need to restart the kernel to use updated packages.
```

Import Dataset

In [36]:
```
data = pd.read_csv('titanic.csv', index_col="PassengerID")
data.head()
```

Out[36]:

PassengerID	Survived	Pass. Class	Name	Age	Nr Siblings/Spouses	Parents/Child
1	0	3	Braund, Mr. Owen Harris	22.0	1	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	
3	1	3	Heikkinen, Miss. Laina	26.0	0	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	
5	0	3	Allen, Mr. William Henry	35.0	0	



In [37]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 1 to 891
Data columns (total 12 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Survived                891 non-null    int64
 1   Pass. Class             891 non-null    int64
 2   Name                    891 non-null    str
 3   Age                     714 non-null    float64
 4   Nr Siblings/Spouses     891 non-null    int64
 5   Nr Parents/Children     891 non-null    int64
 6   Ticket                  891 non-null    str
 7   Fare                    891 non-null    float64
 8   Cabin                   204 non-null    str
 9   Sex                     891 non-null    int64
 10  Embarked - Queenstown   891 non-null    int64
 11  Embarked - Southampton  891 non-null    int64
dtypes: float64(2), int64(7), str(3)
memory usage: 83.7 KB
```

Exploratory Data Analysis

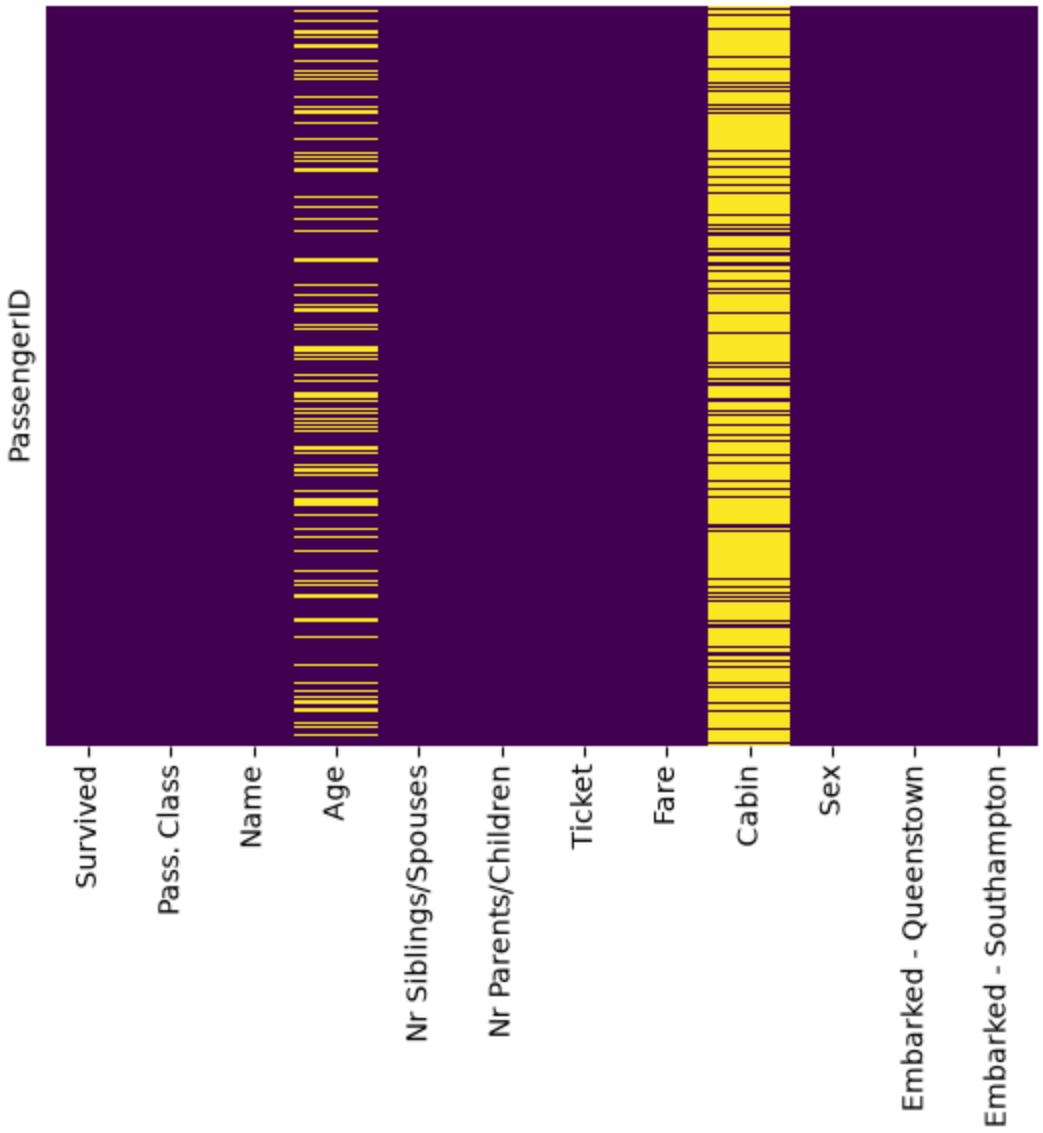
TO DO: Visualize data. Which passenger class has most survivors

Missing Data

We can use seaborn to create a simple heatmap to see where are missing data.

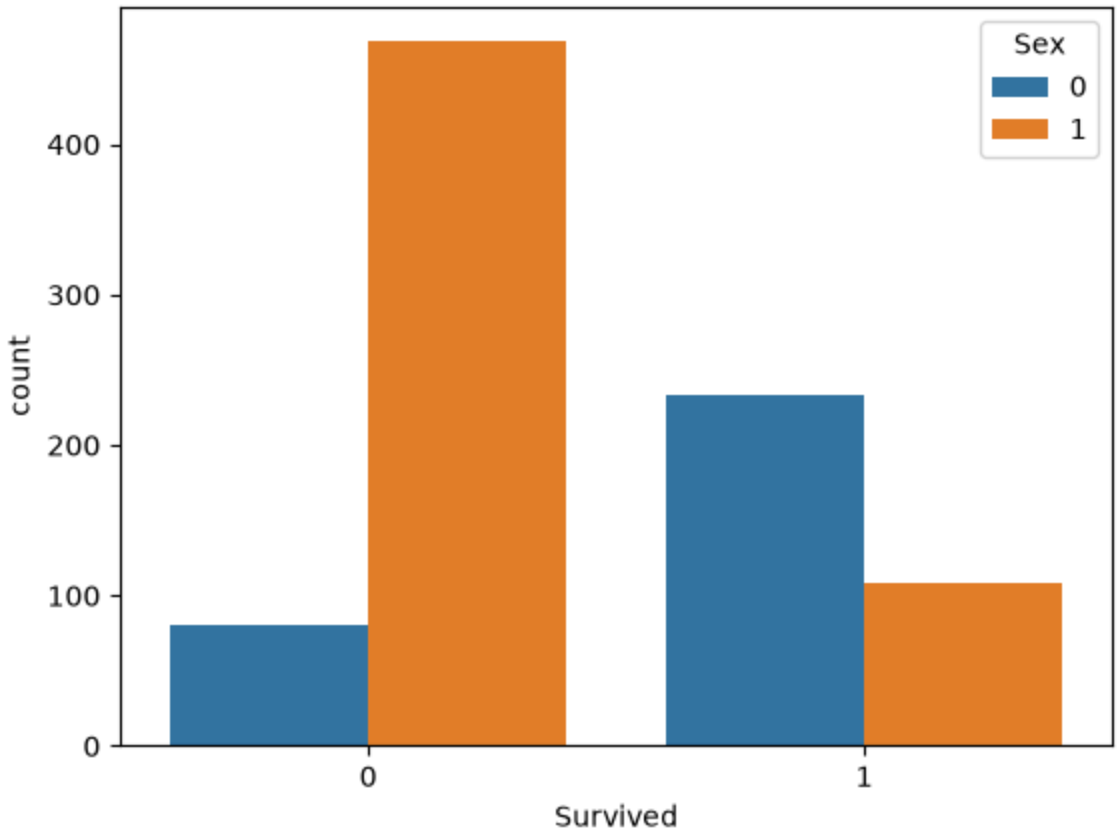
In [38]:
```
sns.heatmap(data.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```

Out[38]:
```
<Axes: ylabel='PassengerID'>
```



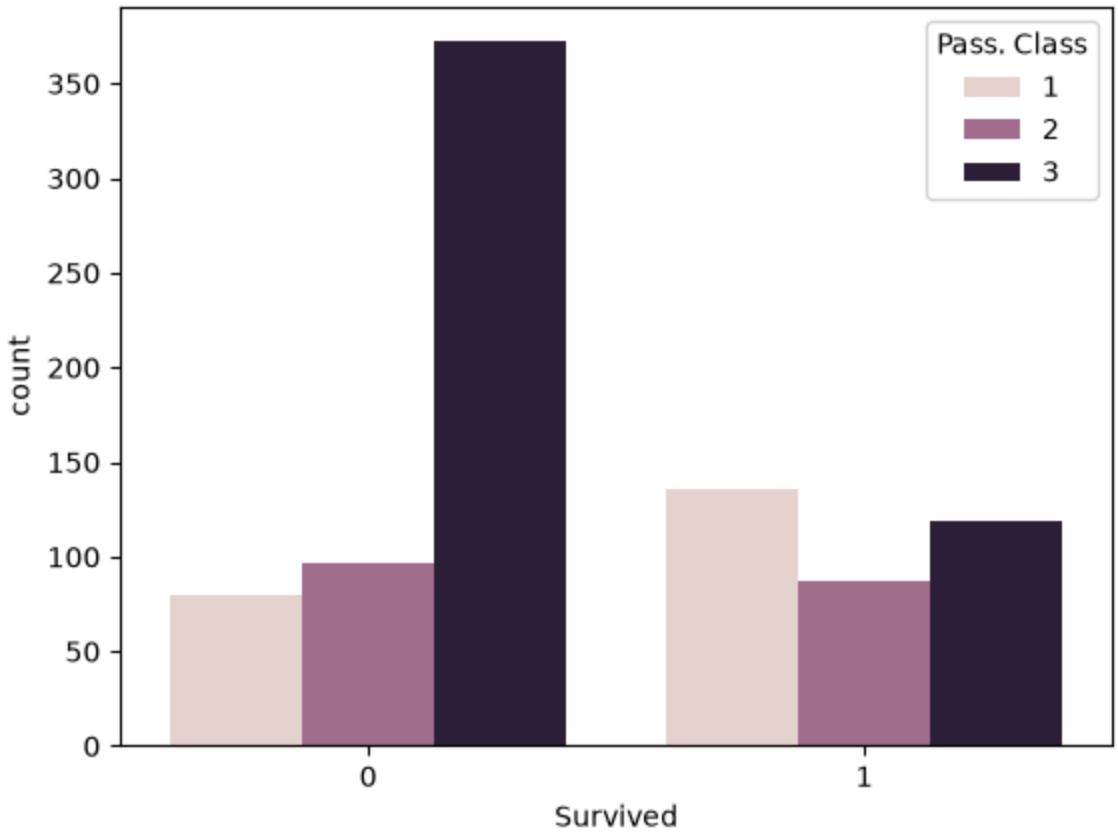
In [39]:
```
sns.countplot(x='Survived', hue='Sex', data=data)
```

Out[39]:
```
<Axes: xlabel='Survived', ylabel='count'>
```



In [40]:
```
sns.countplot(x='Survived', hue='Pass. Class', data=data)
```

Out[40]: <Axes: xlabel='Survived', ylabel='count'>



Missing Data

If data is missing, we cannot use the entire dataset for training. We have to choose:

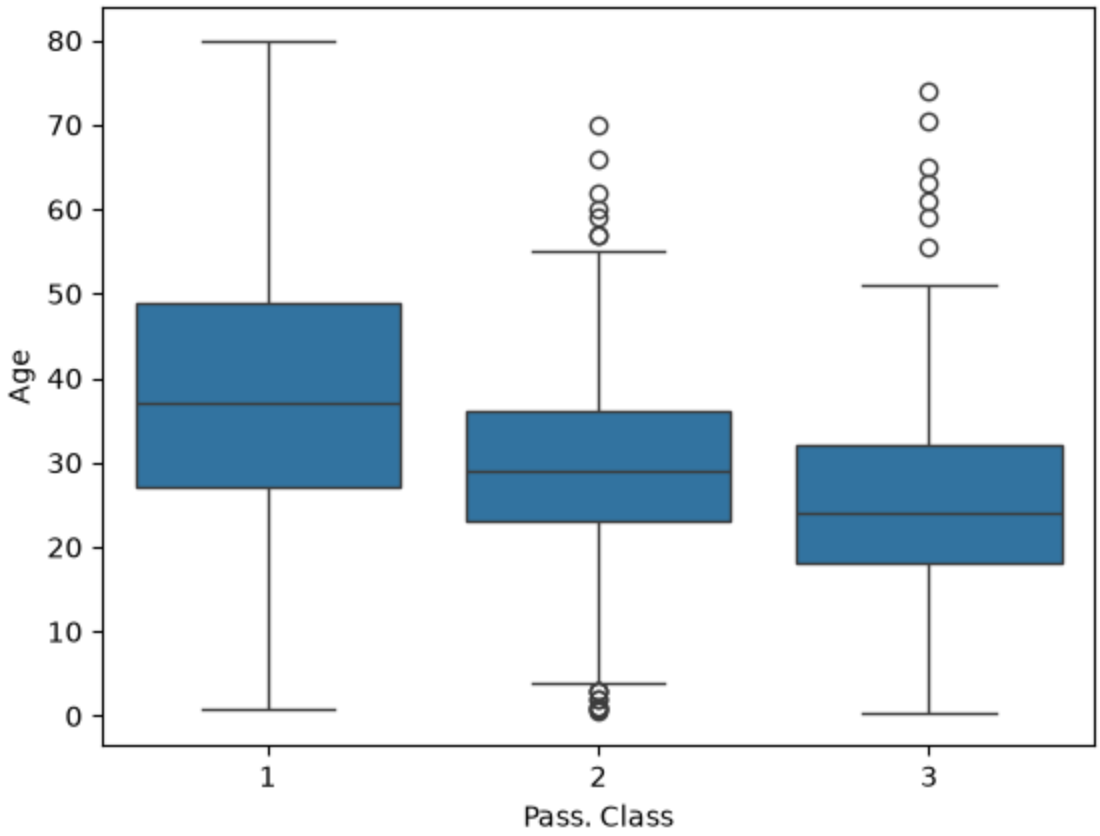
1. Fill the missing data artificially
2. Discard the missing data

TO DO: Visualize the missing data and prepare the dataset for training

Hint: Generate artificial values when the missing data can be estimated and when it is not frequent

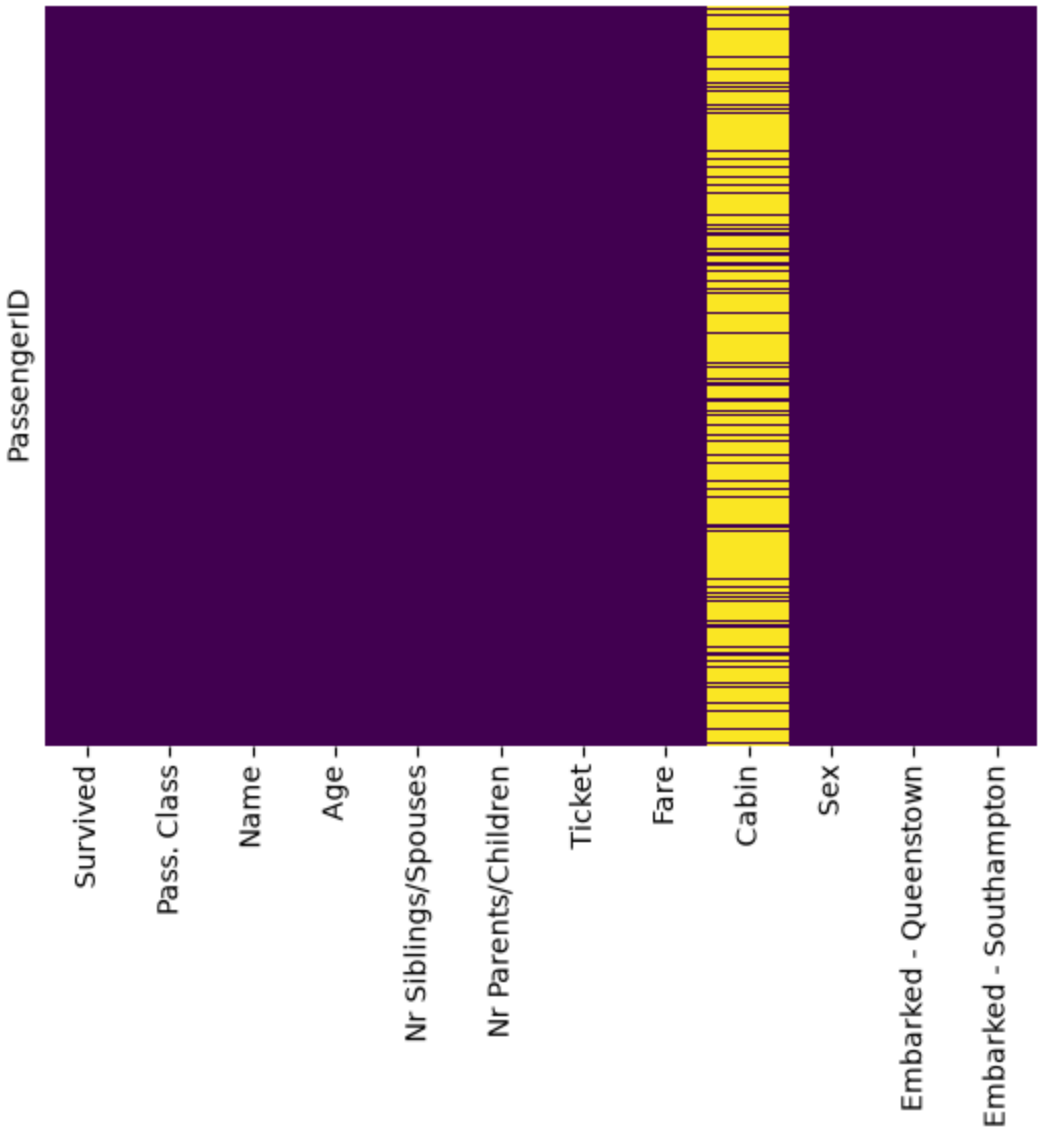
In [41]:
```
sns.boxplot(x='Pass. Class', y='Age', data=data)
# how does age change with the Class (because old people have more money)
```

Out[41]: <Axes: xlabel='Pass. Class', ylabel='Age'>



In [42]:
```
print(data.groupby('Pass. Class')['Age'].median())
```

```
Pass. Class
1    37.0
2    29.0
3    24.0
Name: Age, dtype: float64
```



In [43]:
```
def impute_age(cols):
    Age = cols['Age']
    Pclass = cols['Pass. Class'] # □意: 请确认你的列名就是 'Pass. Class' (中间有空格和点)
    if pd.isnull(Age):
        if Pclass == 1:
            return 37
        elif Pclass == 2:
            return 29
        else:
            return 24
    else:
        return Age
```

In [44]:
```
data['Age'] = data[['Age', 'Pass. Class']].apply(impute_age, axis=1)
sns.heatmap(data.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```

Out[44]: <Axes: ylabel='PassengerID'>

Managing Cabin column

In [45]:
```
data.drop(['Cabin', 'Name', 'Ticket'], axis=1, inplace=True)
```

```
/tmp/ipykernel_20065/434600364.py:1: Pandas4Warning: The inplace keyword in
DataFrame.drop is deprecated and will be removed in a future version (PDEP-
8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inpl
ace for more details.
  data.drop(['Cabin', 'Name', 'Ticket'], axis=1, inplace=True)
```

In [46]:
```
data.head()
```

Out[46]:

PassengerID	Survived	Pass. Class	Age	Nr Siblings/Spouses	Nr Parents/Children	Far
1	0	3	22.0	1	0	7.250
2	1	1	38.0	1	0	71.283
3	1	3	26.0	0	0	7.925
4	1	1	35.0	1	0	53.100
5	0	3	35.0	0	0	8.050

Building a Logistic Regression model

Let's start by splitting our data into a training set and test set as we did in the linear regression case study.

In [47]:
```
from sklearn.model_selection import train_test_split
X = data.drop('Survived', axis=1)
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

TO DO: Train a logistic regression model and use it on the test set

In [48]:
```
from sklearn.linear_model import LogisticRegression
logmodel = LogisticRegression()
logmodel.fit(X_train, y_train)
```

```
/usr/local/python/3.14.2/lib/python3.14/site-packages/sklearn/linear_model/_
logistic.py:599: ConvergenceWarning: lbfgs failed to converge after 100 iter
ation(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regre
ssion
  n_iter_i = _check_optimize_result(
```

Out[48]:

▾ LogisticRegression ⓘ ?

▸ Parameters

▸ Fitted attributes

Performance Evaluation

Confusion Matrix:

$$\begin{matrix} & \text{Observed values} \\ \text{Predicted values} & \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \end{matrix}$$

- **TP (True Positive):** Model predicts *positive* and it's actually *positive*
- **FP (False Positive):** Model predicts *positive* but it's actually *negative* (*false alarm*)
- **TN (True Negative):** Model predicts *negative* and it's actually *negative*
- **FN (False Negative):** Model predicts *negative* but it's actually *positive* (*missed detection*)

F1-score (harmonic mean of precision and recall):

$$F1 = \frac{2 \times precision \times recall}{precision + recall}$$

where:

- precision: $\frac{TP}{TP+FP}$ ratio of true positive (real survivors) over all predicted positive (all predicted as survivors, even if some are not) Precision measures the proportion of positive predictions that were actually correct

- recall: $\frac{TP}{TP+FN}$ ratio of true positive (real survivors) over all that should have been predicted as survivors (all predicted as survivors, even if some are not) Recall measures the proportion of actual positives that were correctly identified.

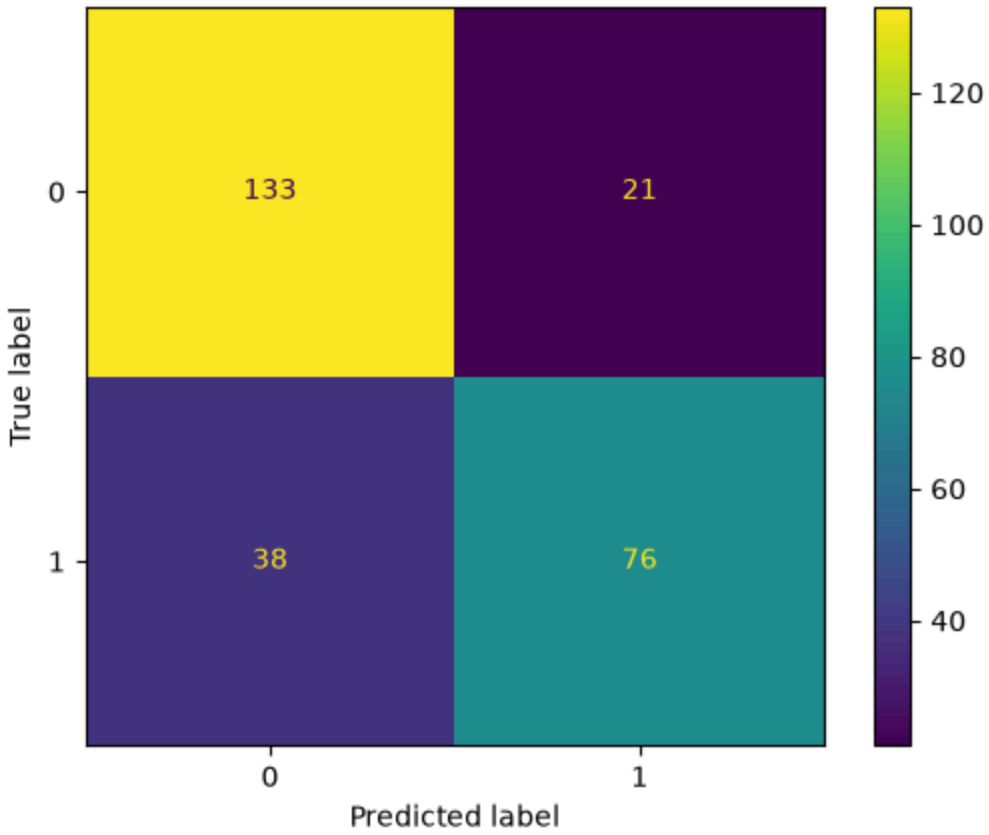
TO DO: Evaluate the performance of the model

In [49]:
```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classi
predictions = logmodel.predict(X_test)
predictions[:10]
```

Out[49]:
```
array([0, 1, 1, 0, 0, 0, 0, 0, 1, 1])
```

In [50]:
```
conf_mat = confusion_matrix(y_test, predictions)
print(conf_mat)

matrix = ConfusionMatrixDisplay(conf_mat).plot(values_format=".0f")
plt.ylabel('True label')
plt.xlabel('Predicted label')
plt.show()
```

```
[[133  21]
 [ 38  76]]
```



In [51]:
```
print(classification_report(y_test, predictions))
```

```
              precision    recall  f1-score   support

           0       0.78      0.86      0.82       154
           1       0.78      0.67      0.72       114

    accuracy                           0.78       268
   macro avg       0.78      0.77      0.77       268
weighted avg       0.78      0.78      0.78       268
```

Not so bad! You might want to explore other features and the full Titanic Kaggle dataset; some suggestions for feature selection:

- Try grabbing the Title (Dr., Mr., Mrs, etc..) from the name as a feature
- Maybe Ticket information could be exploited

Notice that, when dealing with non-numeric features, it is suggested to convert them into numerical values.

Improving the model: non-numerical features

In [52]:
```
# Load original dataset
data = pd.read_csv('titanic.csv', index_col="PassengerID")
data.head()
```

Out[52]:

PassengerID	Survived	Pass. Class	Name	Age	Nr Siblings/Spouses	Parents/Child
1	0	3	Braund, Mr. Owen Harris	22.0	1	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	
3	1	3	Heikkinen, Miss. Laina	26.0	0	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	
5	0	3	Allen, Mr. William Henry	35.0	0	

From non-numerical to numerical features

Regression models use numerical features to estimate an output. Categorical data can be converted to numerical data through one-hot encoding, that is, a binary vector where the value of 1 represents the active category.

Example

Title = {'Mr.', 'Mrs.', 'Miss'}

$$\begin{bmatrix}1\\0\\0\end{bmatrix} \longrightarrow \text{Mr.} \qquad \begin{bmatrix}0\\1\\0\end{bmatrix} \longrightarrow \text{Mrs.} \qquad \begin{bmatrix}0\\0\\1\end{bmatrix} \longrightarrow \text{Miss}$$

TO DO: Convert the "Ticket" and "Title" information into numerical values

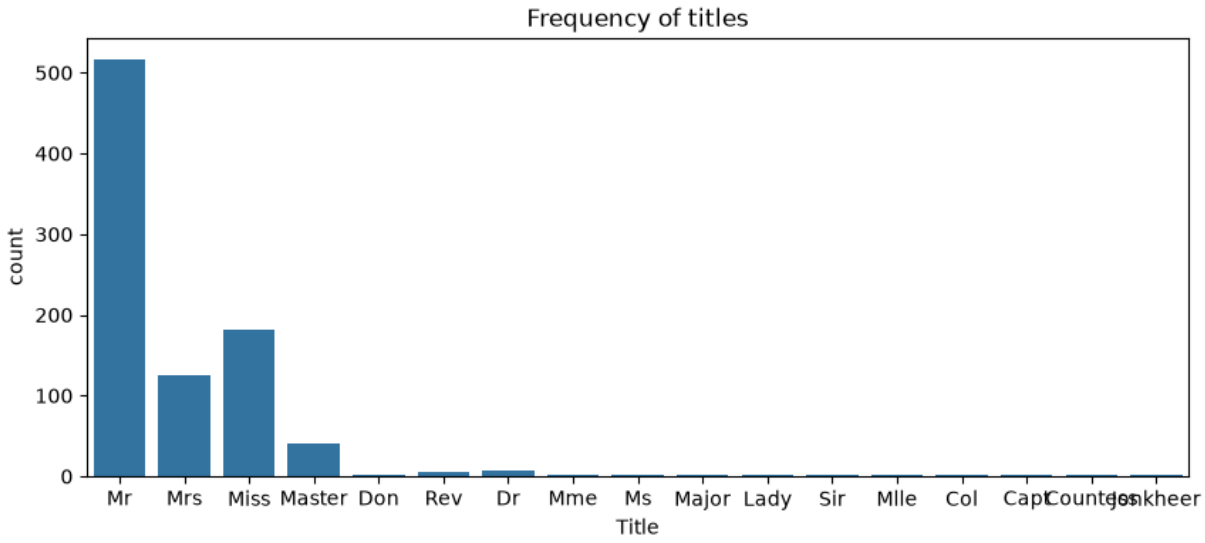
- Hint 1 [see below]: How to extract the Title from the name
- Hint 2 [see below]: Group together different titles to simplify the model
- Hint 3 [see below]: Tickets can be distinguished based on the first letter of their name

In [53]:
```
print(data['Name'].values[0])

# We extract from 'Name' a sequence of letters (A-Za-z) of any length (+) th
data['Title'] = data['Name'].str.extract('([A-Za-z]+)\.')
# data.head()
```

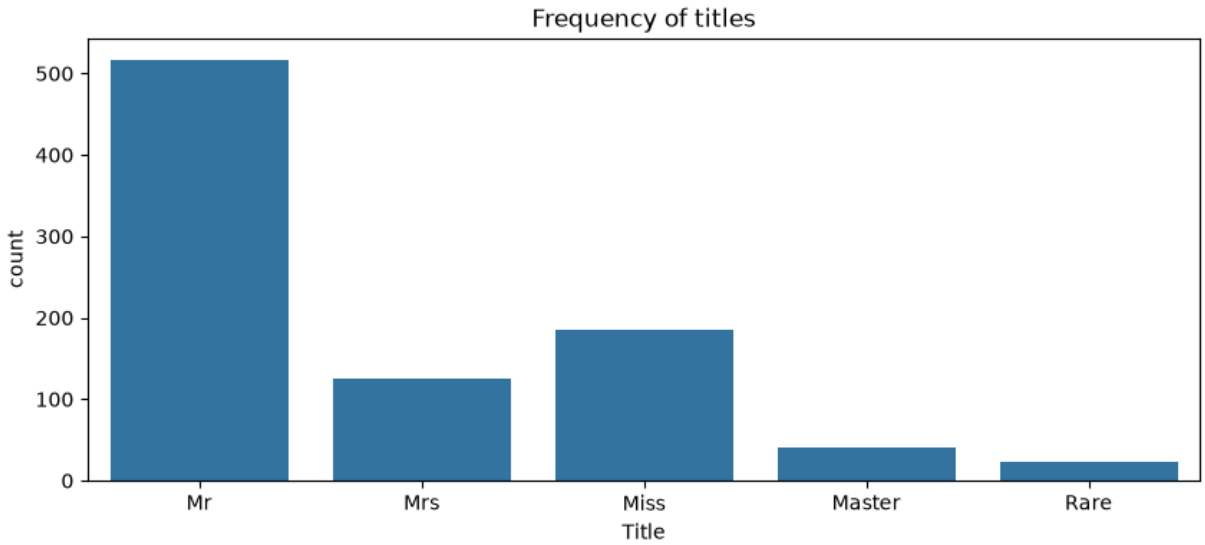
```
Braund, Mr. Owen Harris
<>:4: SyntaxWarning: "\." is an invalid escape sequence. Such sequences will
not work in the future. Did you mean "\\."? A raw string is also an option.
<>:4: SyntaxWarning: "\." is an invalid escape sequence. Such sequences will
not work in the future. Did you mean "\\."? A raw string is also an option.
/tmp/ipykernel_20065/590210649.py:4: SyntaxWarning: "\." is an invalid escap
e sequence. Such sequences will not work in the future. Did you mean "\\."?
A raw string is also an option.
  data['Title'] = data['Name'].str.extract('([A-Za-z]+)\.')
```

In [ ]:
```
print(data['Title'].unique())

plt.figure(figsize=(10, 4))
sns.countplot(data=data, x='Title')
plt.title('Frequency of titles')
plt.show()
# it's not good to have many categories with few data for that category
```

```
<StringArray>
[      'Mr',      'Mrs',     'Miss',   'Master',      'Don',      'Rev',
       'Dr',      'Mme',       'Ms',    'Major',     'Lady',      'Sir',
     'Mlle',      'Col',     'Capt', 'Countess', 'Jonkheer']
Length: 17, dtype: str
```



In [55]:
```
# We group together different classes
data['Title'] = data['Title'].replace('Mme', 'Mrs')
data['Title'] = data['Title'].replace(['Mlle', 'Ms'], 'Miss')
data['Title'] = data['Title'].replace(['Lady', 'Countess', 'Capt', 'Col', 'D

plt.figure(figsize=(10, 4))
sns.countplot(data=data, x='Title')
plt.title('Frequency of titles')
plt.show()
```



In [ ]:
```
# We want to extract the first letter of the Ticket (given information)
data['Ticket_prefix'] = data['Ticket'].apply(lambda x: x.split()[0] if not x
data['Ticket_prefix'][:5]

# or
# def split_if_not_digit(x):
#     if not x.isdigit():
#         return x.split()[0]
#     return 'No_prefix'

# lambda x: x.split()[0] if not x.isdigit() else 'No_prefix'
```

Out[ ]:
```
PassengerID
1          A/5
2           PC
3     STON/O2.
4    No_prefix
5    No_prefix
Name: Ticket_prefix, dtype: str
```

In [57]:
```
data = pd.get_dummies(data, columns=['Ticket_prefix'], drop_first=True)
data = pd.get_dummies(data, columns=['Title'], drop_first=True)
data.head()
```

Out[57]:

PassengerID	Survived	Pass. Class	Name	Age	Nr Siblings/Spouses	Parents/Child
1	0	3	Braund, Mr. Owen Harris	22.0	1	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	
3	1	3	Heikkinen, Miss. Laina	26.0	0	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	
5	0	3	Allen, Mr. William Henry	35.0	0	

5 rows × 59 columns

Re-train

Drop original 'Name' and 'Ticket' columns as we've extracted relevant info and estimate age as before

In [58]:
```
data.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)
data['Age'] = data[['Age','Pass. Class']].apply(impute_age, axis=1)
```

```
/tmp/ipykernel_20065/3771052950.py:1: Pandas4Warning: The inplace keyword in
DataFrame.drop is deprecated and will be removed in a future version (PDEP-
8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inpl
ace for more details.
  data.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)
```

Train-test split with new features, train new model, and evaluate performance

In [59]:
```
# Data splitting
X = data.drop('Survived', axis=1)
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra

# Train model
logmodel = LogisticRegression(max_iter=1000)
logmodel.fit(X_train, y_train)

# Predict
predictions = logmodel.predict(X_test)

# Performance
print(classification_report(y_test, predictions))
```

```
              precision    recall  f1-score   support

           0       0.80      0.90      0.85       154
           1       0.84      0.70      0.77       114

    accuracy                           0.82       268
   macro avg       0.82      0.80      0.81       268
weighted avg       0.82      0.82      0.81       268
```

AS WE CAN SEE WE GOT AN IMPROVEMENT!!!

Predictive Imputation

Instead of estimating the age with the median value, why don't we train a regression model?

In [60]:
```
# Import original dataset and keep only rows relevant for age prediction
data = pd.read_csv('titanic.csv', index_col="PassengerID")
age_df = data[['Age', 'Pass. Class', 'Nr Siblings/Spouses', 'Nr Parents/Chil
```

TO DO: Build a linear regression model to estimate the missing age

In [61]:
```
# Data splitting
age_notnull = age_df[age_df['Age'].notnull()]
age_null = age_df[age_df['Age'].isnull()]
```

In [62]:
```
# Data splitting
X_age_train = age_notnull.drop('Age', axis=1)
y_age_train = age_notnull['Age']

# Not really a test data
X_age_test = age_null.drop('Age', axis=1)
```

In [63]:
```
from sklearn.linear_model import LinearRegression

linreg = LinearRegression()
linreg.fit(X_age_train, y_age_train)

predicted_ages = linreg.predict(X_age_test)
data.loc[data['Age'].isnull(), 'Age'] = predicted_ages
```

Perform logistic regression training as before

In [64]:
```
# Remove non-numerical values
data.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)

# Train-test split
X = data.drop('Survived', axis=1)
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra

# Train model
logmodel = LogisticRegression(max_iter=1000)
logmodel.fit(X_train, y_train)

# Predict and evaluate
predictions = logmodel.predict(X_test)
print(classification_report(y_test, predictions))
```

```
              precision    recall  f1-score   support

           0       0.78      0.88      0.83       154
           1       0.80      0.68      0.73       114

    accuracy                           0.79       268
   macro avg       0.79      0.78      0.78       268
weighted avg       0.79      0.79      0.79       268
```

```
/tmp/ipykernel_20065/1376050733.py:2: Pandas4Warning: The inplace keyword in
DataFrame.drop is deprecated and will be removed in a future version (PDEP-
8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inpl
ace for more details.
  data.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)
```

We got a little improvement

You can try to combine the linear regression imputation and the non-numerical features